UNIVERSITY OF GHANA

LIFEGUARD:

WEARABLE HEALTH AND ENVIRONMENTAL MONITORING SYSTEM

FINAL DEFENSE

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INTRODUCTION

- Current health and environmental monitoring systems require multiple devices, making them costly, complex, and less accessible. Many also lack real-time alerts and comprehensive data, leaving gaps in timely intervention.

- LifeGuard, powered by the advanced **Nicla Sense ME** board, integrates nine sensors to deliver seamless real-time monitoring of health metrics and environmental conditions. The system employs sophisticated motion detection algorithms and environmental sensing, creating a holistic safety monitoring solution for proactive risk identification.

- With features like fall detection, physical activity recognition, environmental condition assessment, and air quality analysis, LifeGuard represents a cost-effective and integrated solution for personal protection and well-being.

PROBLEM DEFINITION

- Most market solutions rely on multiple devices for health and environmental monitoring, leading to higher costs and added complexity.

- Need for comprehensive safety monitoring for vulnerable populations. This ensures proactive identification of risks and timely intervention to safeguard their well-being.

- Many existing solutions fail to provide real-time alerts and updates, limiting their ability to respond promptly to critical situations.

- Growing demand for preventive healthcare technology. This shift highlights the importance of innovative solutions that empower individuals to monitor and manage their health before conditions escalate.

RELEVANCE OF WORK

- **Aging Population**

WHO reports **28-35%** of people aged 65+ fall each year, highlighting the urgent need for advanced fall detection systems and general health monitoring to reduce injuries, hospitalizations, and improve quality of life for the elderly.

- **Environmental Concerns**

Environmental health concerns, especially air quality, affect public health, with **99%** of the global population exposed to unsafe pollution levels, leading to over 7 million premature deaths annually (World Health Organization, Environmental Health Impact Report, October 2024).

RELEVANCE OF WORK

- **Healthcare Costs**

Cost-effective solutions needed to make safety monitoring accessible to broader populations. This will help bridge the gap in safety monitoring, ensuring equitable protection for underserved communities.

- **Industrial Safety**

Rising industrial accidents due to environmental hazards necessitate real-time monitoring solutions. Proactive detection and response systems can significantly reduce risks, safeguarding both workers and assets.

AIMS & OBJECTIVES

- **Integrated Monitoring System**

Develop an integrated health and environmental monitoring system

- **Fall Detection and Activity Recognition**

Implement real-time fall detection and activity recognition

- **Environmental Condition Monitoring**

Create comprehensive environmental condition monitoring

- **Alert System**

Establish efficient alert system for emergency situations

- **User Interfaces**

Design user-friendly mobile and web interfaces

LITERATURE REVIEW/EXISTING WORKS

Author(s)	Title	Achieved	Gap	Lifeguard’s Solution
D. Hemapriya; Pavithra Viswanath; V. M. Mithra; [2019]	Wearable medical devices: Design challenges and issues	Identified key challenges in wearable device development	Limited focus on environmental monitoring integration	Integrates air quality, humidity, and temperature sensors with health metrics for holistic safety.
Mojtaba Hosseininejad, Sajad Ayoobi Yazdi, et. [2021]	Wearable Devices in Health Monitoring from the Environmental towards Multiple Domains: A Survey	Analyzed sensor fusion for health/environmental data integration	Limited focus on real-time data fusion and actionable insights.	LifeGuard employs ML-driven anomaly detection (e.g., LSTM) to convert raw data into real-time alerts and responses

LITERATURE REVIEW/EXISTING WORKS

Author(s)	**Title**	**Achieved**	**Gap**	**Lifeguard's Solution**
Tong Zhu, Shunqing Xu et al. [2024]	Advances and Perspectives in Environmental Health Research in China	Linked pollution exposure to health outcomes (e.g., preterm births, cardiovascular risks)	Limited real-time monitoring tools for personalized exposure assessment	LifeGuard's portable sensors provide real-time air quality alerts (CO2, VOCs) and correlate data with health metrics for preventive care
Ali Chelli, Matthias Patzold [2019]	A Machine Learning Approach For Fall Detection and Daily Living Activity Recognition	Validated AI models for specific use cases like fall detection	No combined analysis with health metrics such as heart rate variability	Correlates accelerometer data with Oxygen Saturation & Pulse(e.g., distinguishing falls from jumps)

RESEARCH/DESIGN METHODOLOGY

Approach

- Agile development with iterative prototyping (2-week sprints)

Hardware Setup

- Sensor Configuration
  - Configure Nicla Sense ME board via BLE and MAX30102 sensor for pulse.
  - Optimize sensor fusion (IMU + environmental data) for combined motion/air quality analysis
- Power Management
  - Use LiPo(3.7V, 400mAh) battery with low-power sleep modes for 72h operation.

RESEARCH/DESIGN METHODOLOGY

Machine Learning Pipeline

- Data Collection
  - Gather labeled datasets for activities (walking, falls) and environmental thresholds (e.g., CO_2 > 1000ppm).

- Preprocessing
  - Processed accelerator values & gyroscopic values

- Model Training (Pre-Trained TinyML Models)
  - Utilized custom trained model (LSTM)

- Deployment
  - Deployed model as an Arduino library for Nicla’s 32-bit Cortex-M4 microcontroller to enable real-time alerts and edge inference.

RESEARCH/DESIGN METHODOLOGY

Backend Integration

- Custom .NET Backend
  - RESTful APIs for data ingestion from Nicla Sense ME.
  - PostgreSQL database with HIPAA-compliant encryption for health and environmental data.
  - Integration with React and Flutter frontend for real-time dashboards.
- Alert System
  - Threshold-based triggers (e.g., fall → SMS to emergency contacts).
  - Pollution map overlays using Mapbox API.

RESEARCH/DESIGN METHODOLOGY

Testing & Validation

- Performance Metrics
  - Accuracy (precision/recall for fall detection: >95%).
  - Latency (<500ms end-to-end alert delivery).
  - Power consumption (<10mA average draw).
- Field Testing
  - Iterate based on false-positive rates and usability feedback.

WORKING SYSTEM OVERVIEW

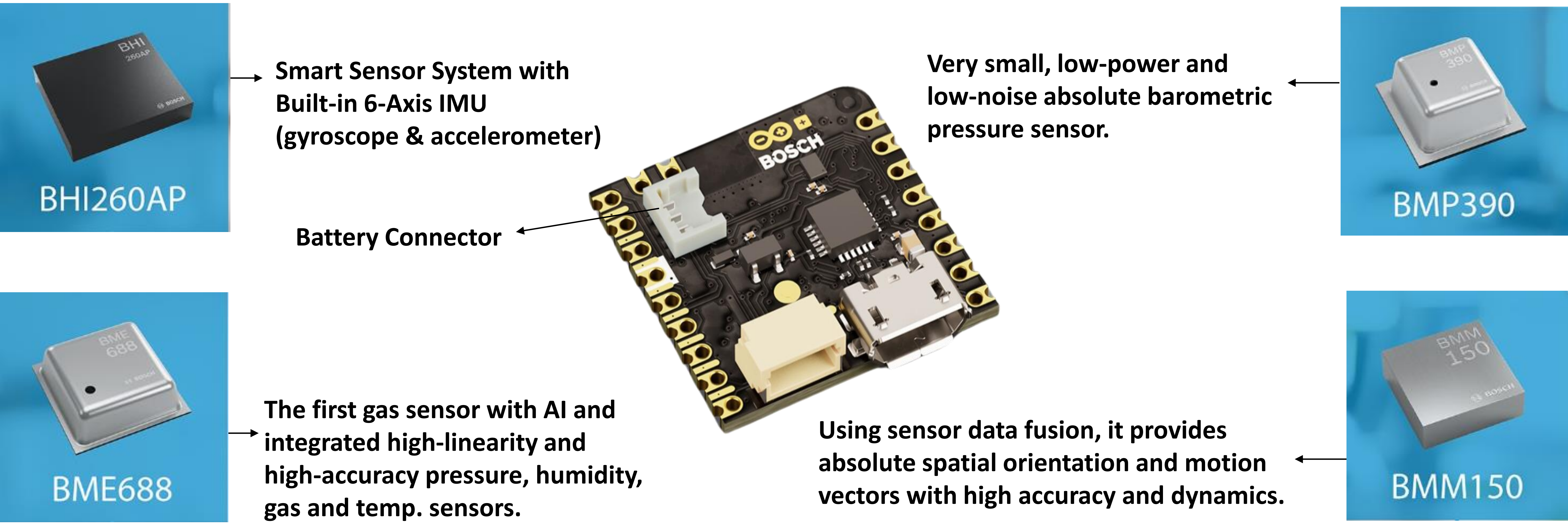


Figure 1: Working System Overview

PHYSICAL HARDWARE ASSEMBLY

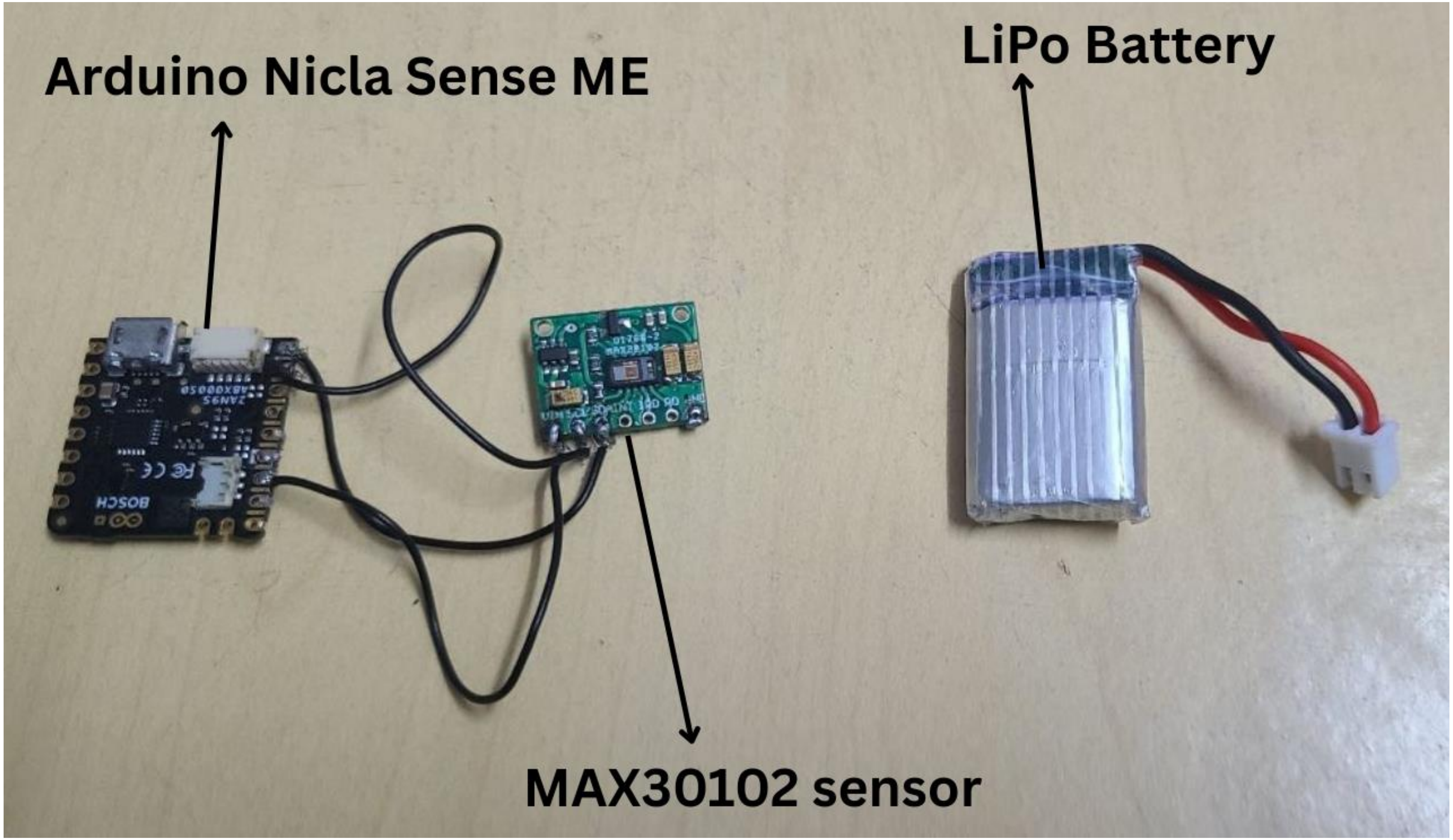
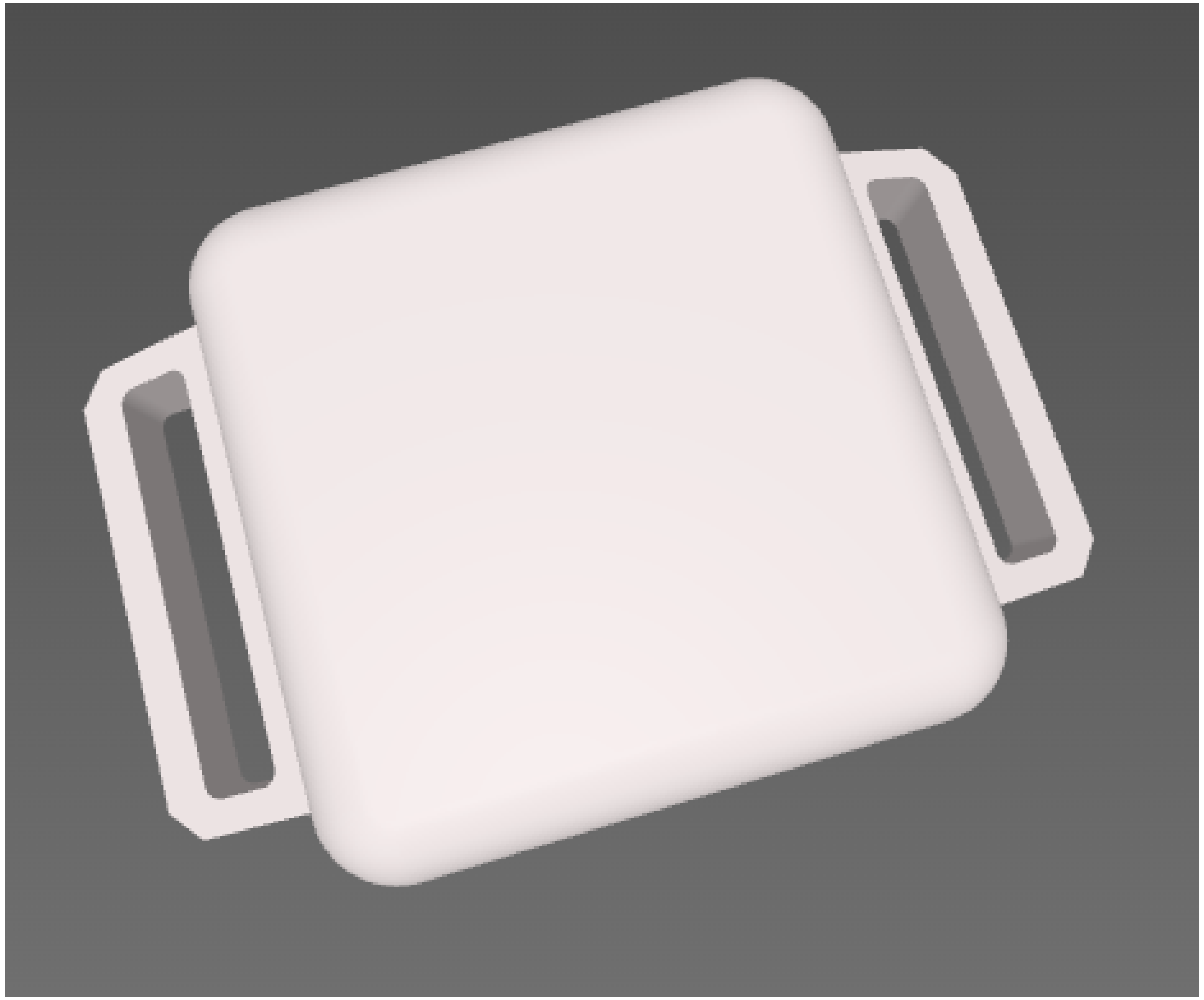
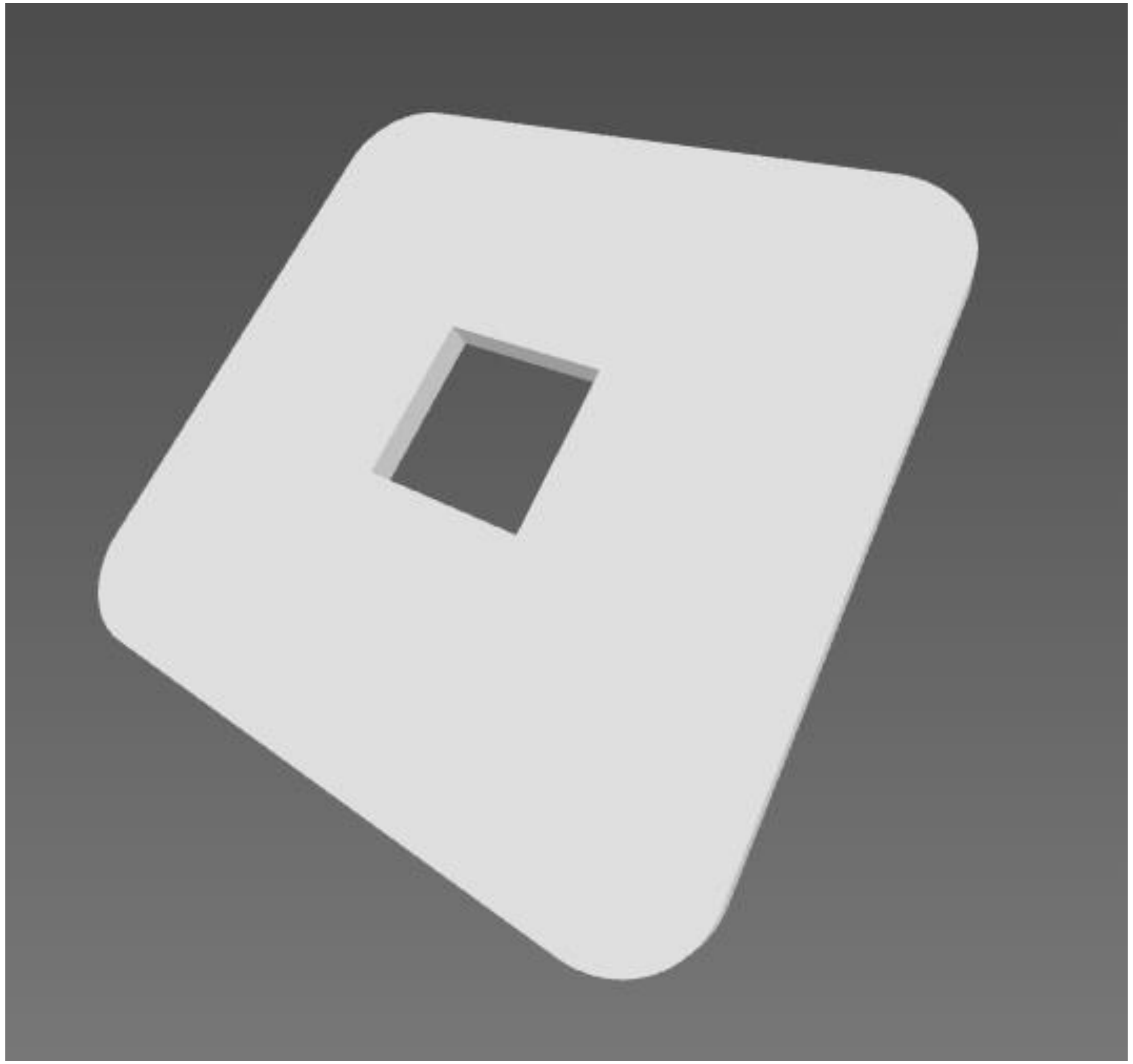


Figure 5: Physical Hardware Assembly

SOLIDWORKS DESIGNS FOR DEVICE FRAME

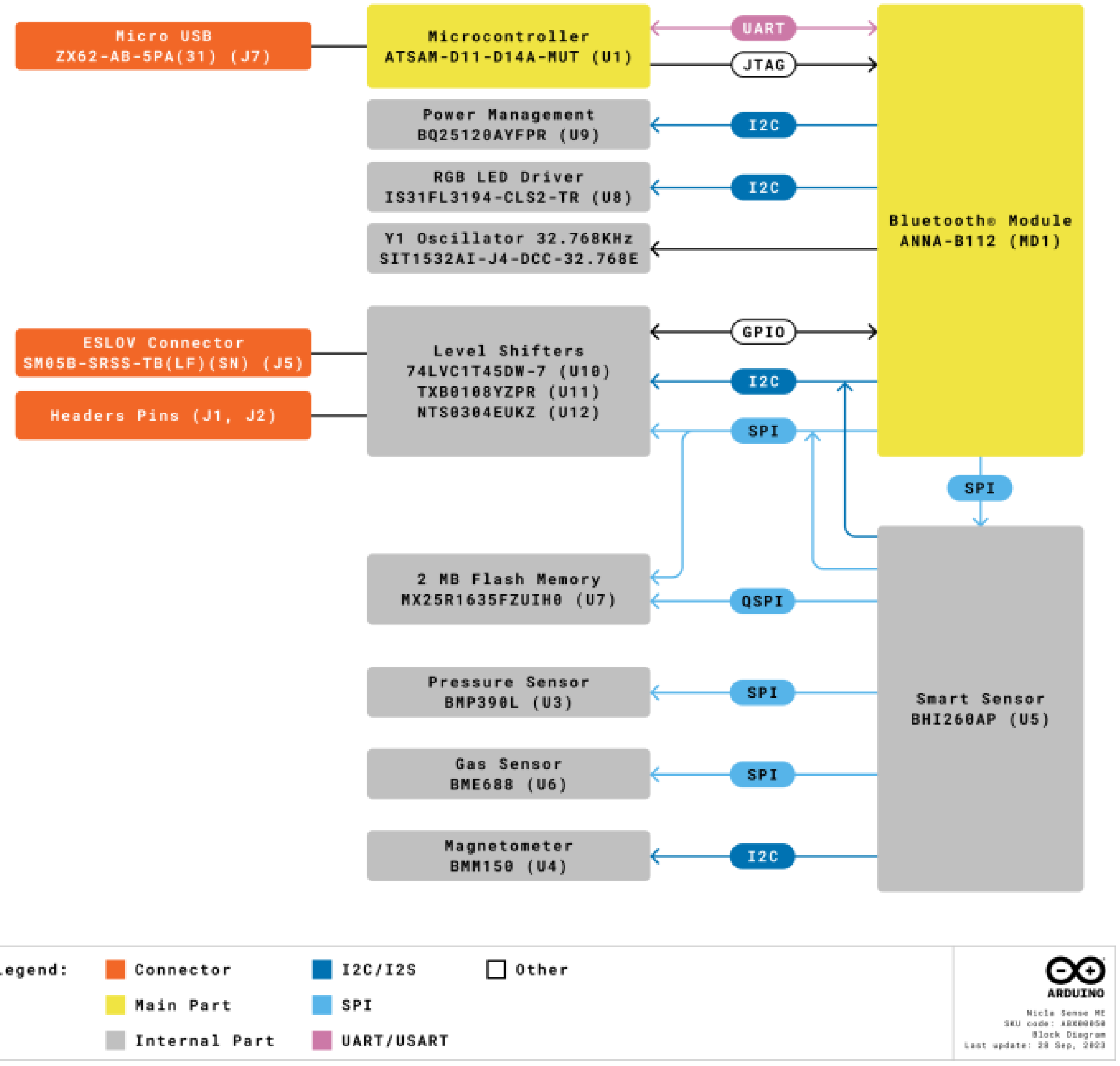




FINAL DEVICE DESIGN

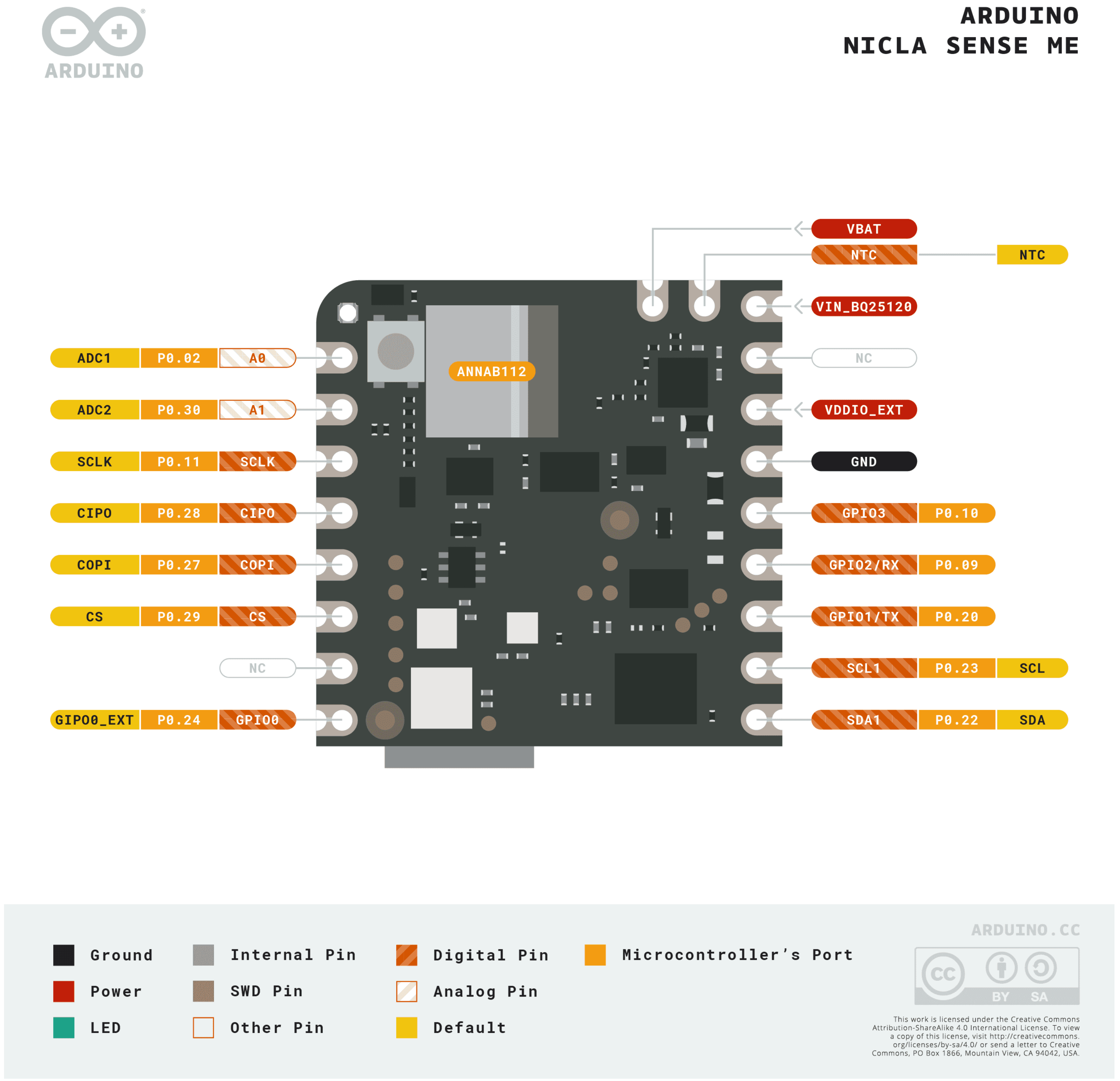
Figure 6: Final Device Design

NICLA SENSE ME BLOCK DIAGRAM

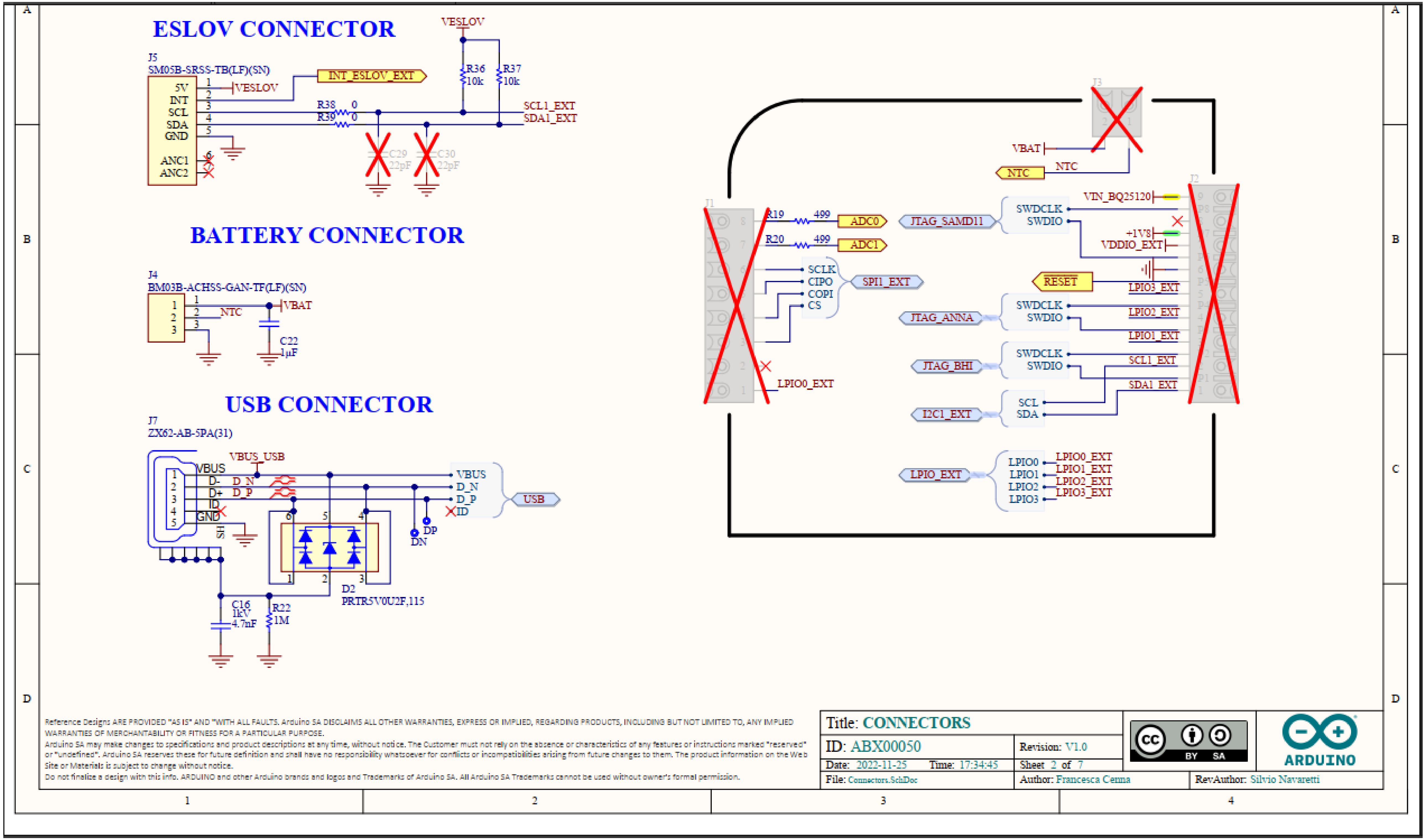


Nicla Sense ME Block Diagram

ARDUINO NICLA SENSE ME PINOUT CONFIGURATION



HARDWARE IMPLEMENTATION & SCHEMATICS



SYSTEM WIRING DIAGRAM

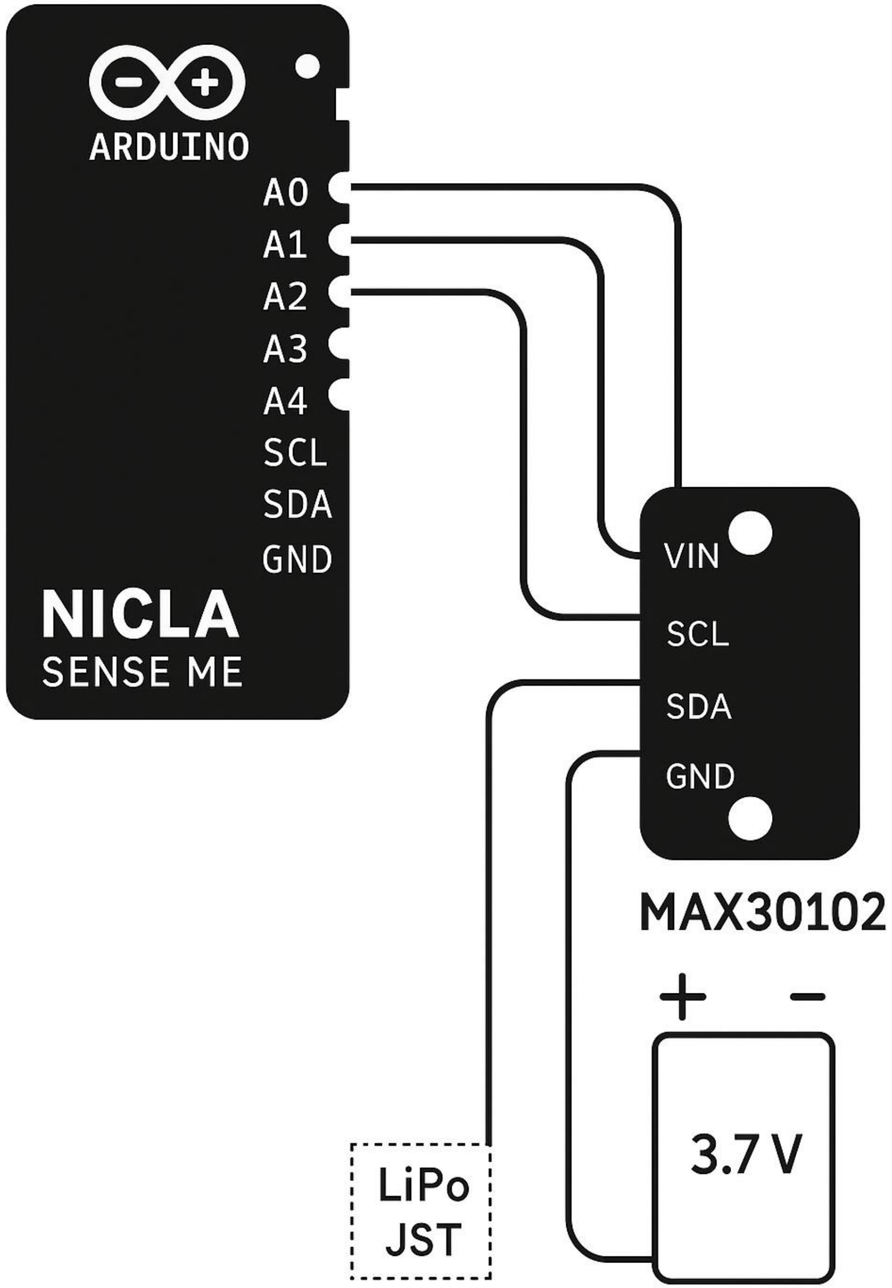


Figure 5: System Wiring Diagram

SYSTEM ARCHITECTURE

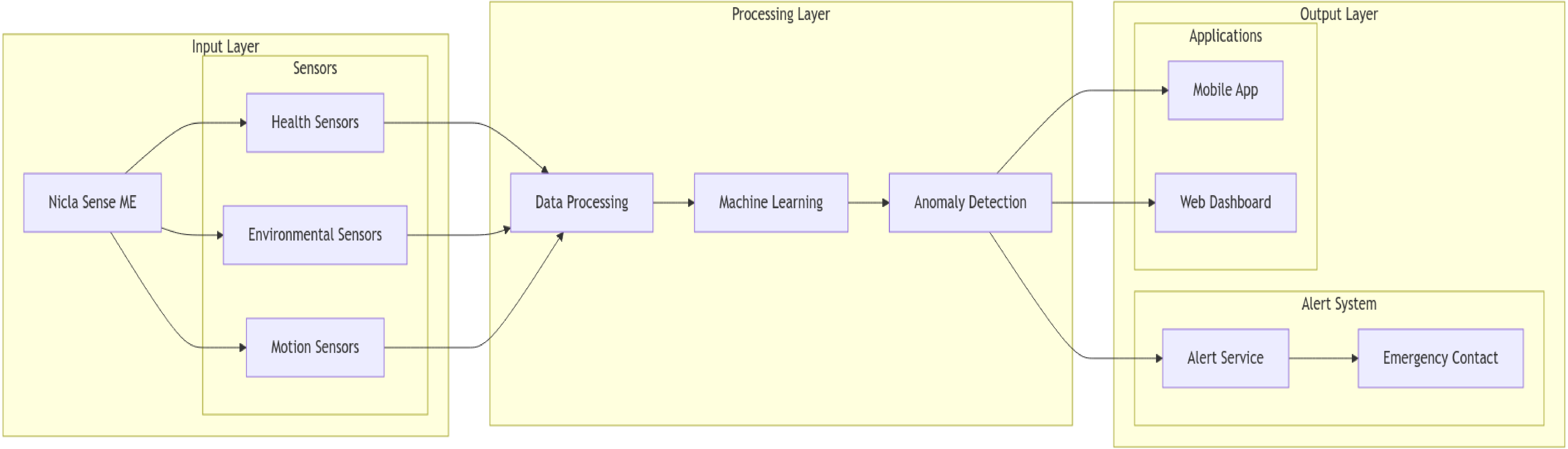


Figure 2: Proposed System Architecture

DATA FLOW DIAGRAM

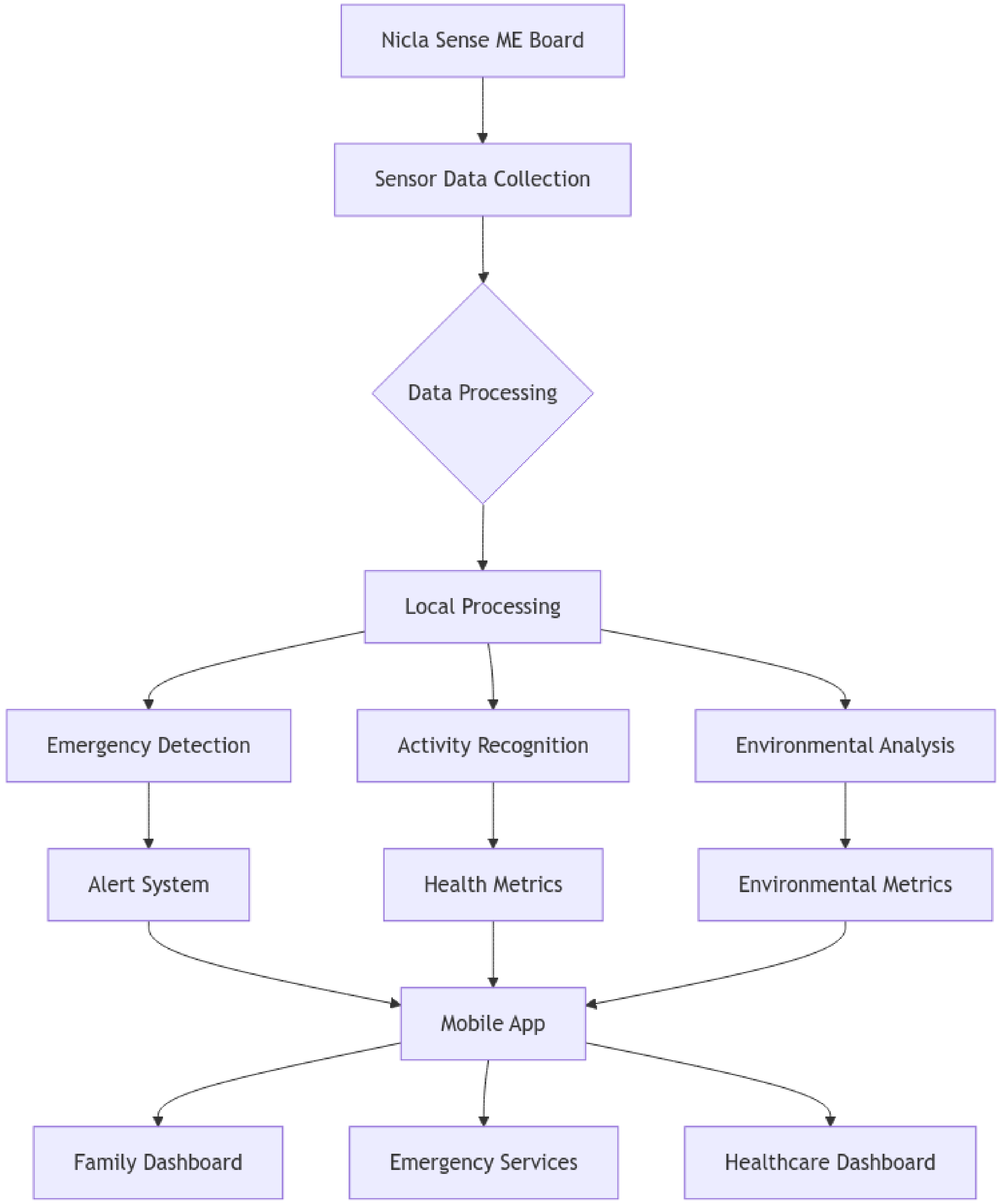


Figure 3: Proposed Data Flow Diagram

PICTORIAL VIEW OF SYSTEM OVERVIEW

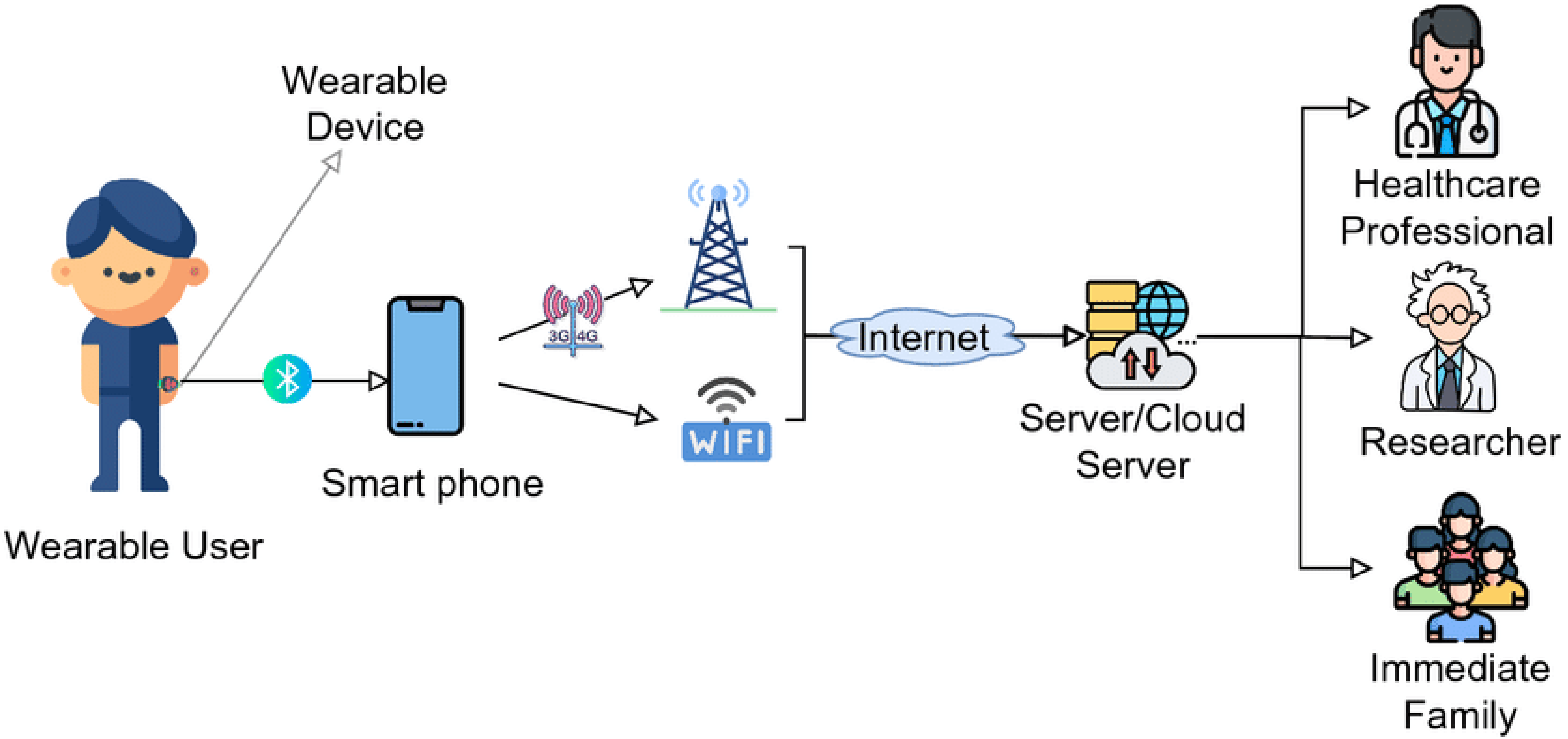


Figure 4: Pictorial view of proposed system overview

SYSTEM REQUIREMENTS

Functional Requirements

- Real-time health monitoring
  - Fall detection
  - Activity recognition
  - Heart rate variability
- Environmental monitoring
  - Temperature sensing
  - Humidity monitoring
  - Air quality analysis
- Alert system
  - Emergency notifications
  - Automated emergency contact
  - Custom alert thresholds

Non-Functional Requirements

- Performance
  - 24/7 operation capability
  - 99.9% system uptime
- Security
  - End-to-end encryption
  - HIPAA compliance
  - Secure data storage
- Usability
  - Intuitive interface
  - Long lasting Battery life
  - Water-resistant design

TECHNOLOGIES USED

Core Hardware Components

- Arduino Nicla Sense ME board
- MAX30102 Sensor
- LiPo battery(3.7V, 400mAh)
- Custom-designed enclosure

Sensors (built-in)

- Accelerometer & Gyroscope (Motion detection)
- Temperature & Humidity sensors
- Barometric pressure sensor
- Magnetometer

TECHNOLOGIES USED

Software Stack

Frontend Technologies

- React (Web Dashboard)
- Flutter (Mobile App)
- TypeScript
- Tailwind CSS

Backend Technologies

- Firebase (Real time database)
- Node JS
- .NET
- PostgreSQL

Firmware

- Platform IO
- C++ (Arduino Framework)

TECHNOLOGIES USED

Hosting Services

- Vercel/Render (Web Hosting)
- Render (Backend Hosting)
- Neon (Database Hosting)

Authentication & Security

- Google Authentication
- Oauth 2.0
- End-to-end encryption
- JWT Authentication

Machine Learning & Analytics

- Quantization and pruning for real-time edge inference
- LSTM networks for fall detection
- Sensor data normalization (z-score).
- Time-series windowing for motion data.
- Pollution map & analytics using Mapbox

REAL WORLD APPLICATIONS

Healthcare

- Remote Patient Monitoring
- Early warning system for respiratory issues
- Fall prevention for elderly
- Emergency response automation

Industrial Safety

- Worker safety in hazardous environments
- Air quality monitoring in confined spaces
- Air Quality triggers

REAL WORLD APPLICATIONS

Personal Wellness

- Environmental impact on health tracking
- Exercise condition monitoring such as Walking, Running, Jumping, Sitting, etc.
- Indoor air quality alerts for gases such as Volatile Organic Compounds (VOCs), Volatile Sulfur Compounds (VSCs), Carbon Monoxide and Hydrogen in the ppb range.
- Weather-related health warnings
- Real-time location
- Emergency contact and alert system

Outdoor Recreation

- Adventure sports activity tracking
- Environmental condition alerts
- Weather hazard warnings

TECHNICAL CHALLENGES

Power Management & Battery Life

- Optimizing power consumption for continuous 24/7 monitoring using a stable rechargeable battery
- Balancing sensor sampling rates with battery efficiency
- Implementing effective power-saving modes without compromising functionality

Real-time Processing & Communication

- Implementing efficient data processing at the edge
- Managing continuous data streaming over BLE
- Ensuring reliable wireless connectivity in various environments
- Handling potential network latency and disconnections

ROLES & RESPONSIBILITIES

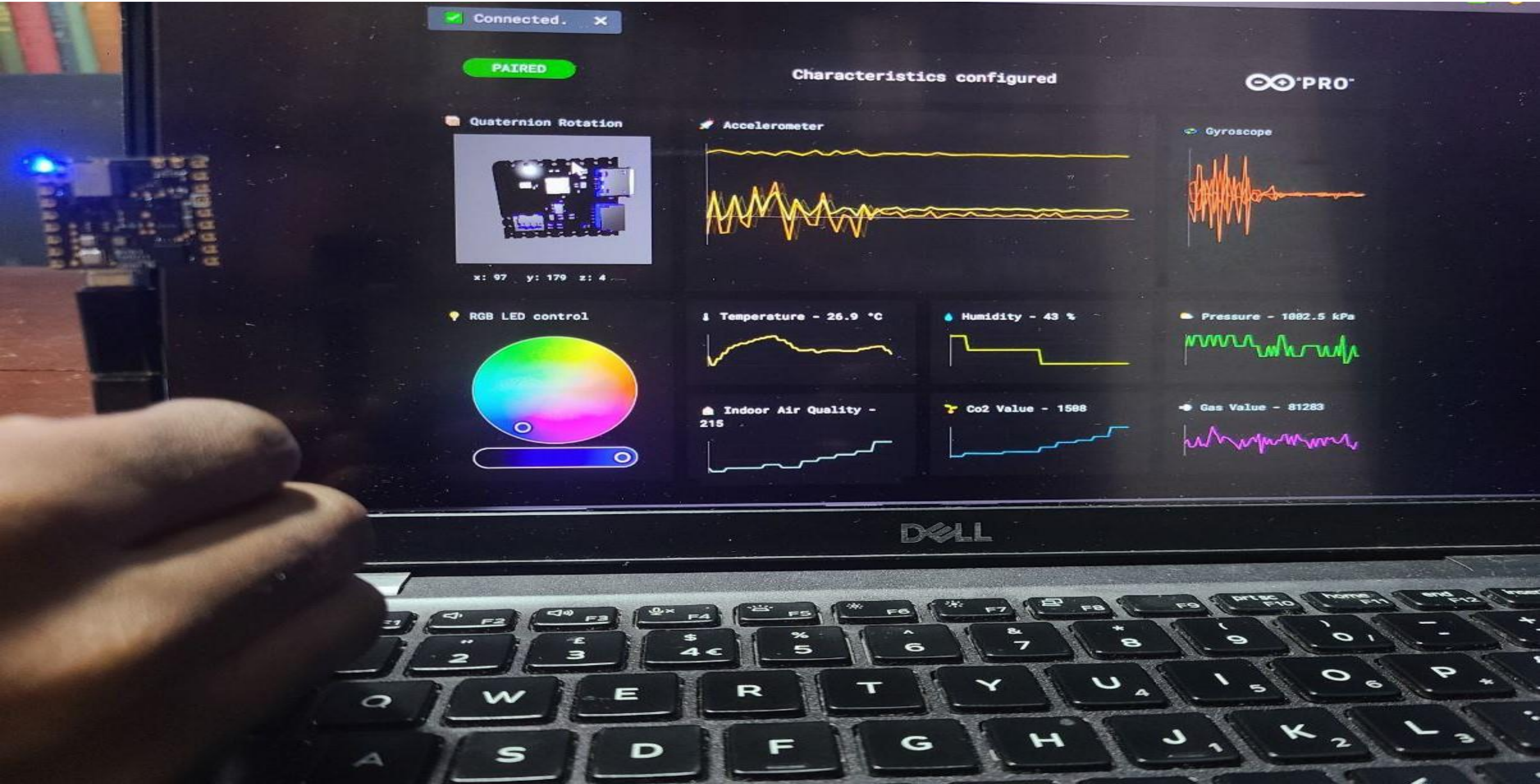
Team Member	Key Focus Areas
Evans Acheampong	Full stack development (React, Flutter, Node), hardware and sensor integration, and user interface testing/documentation.
Michael Adu-Gyamfi	Backend development (.NET, PostgreSQL), machine learning, firmware, data analytics and system security (encryption, CI/CD), Real time database (firebase).

BILL OF MATERIALS

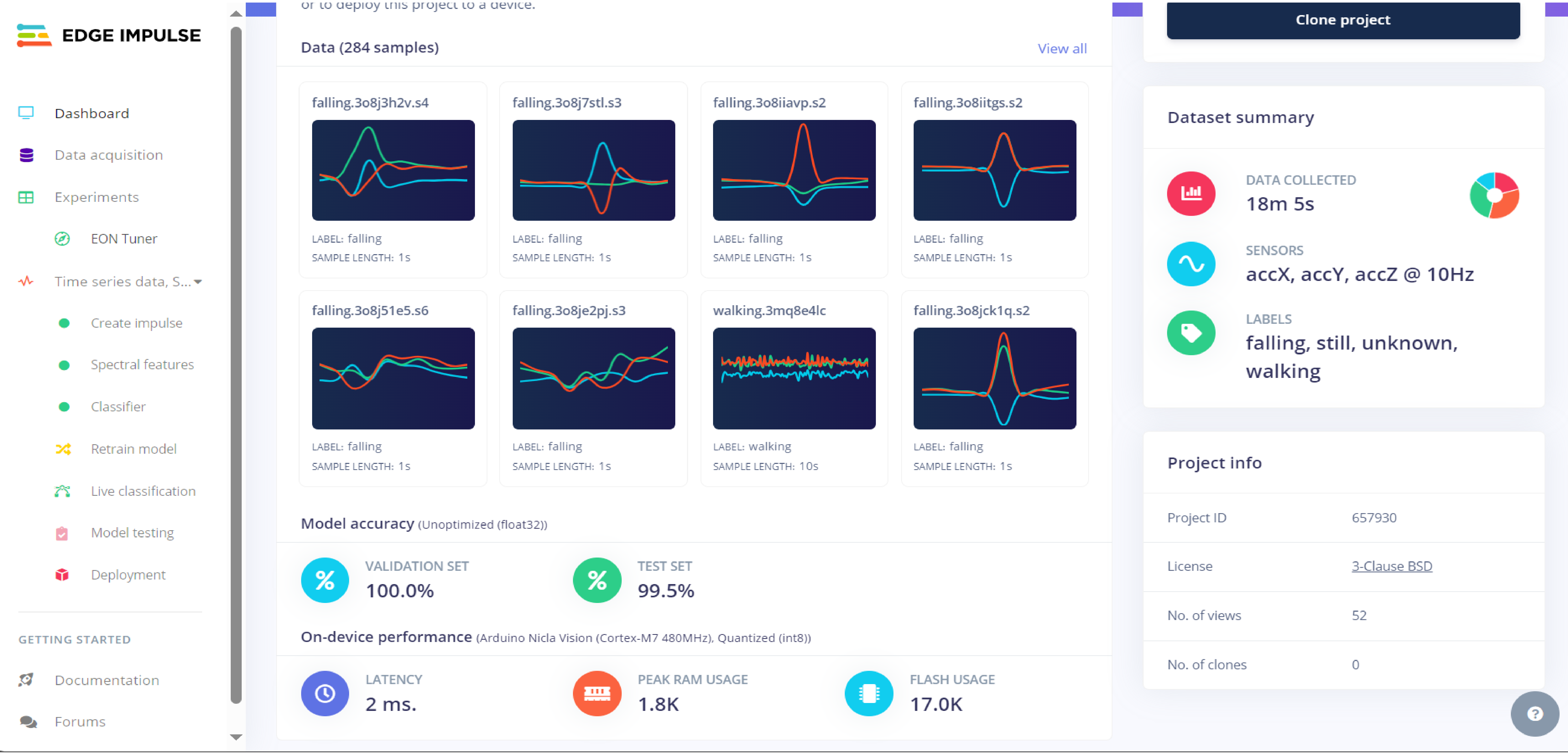
Item	Cost (GH₵)	Quantity	Total (GH₵)
Arduino Nicla Sense ME	1500.00	1	1500
MAX30102 Sensor – Heart Rate & Pulse Oximeter Sensor Module	64.00	1	64.00
LiPo Battery – 3.7V	95.00	1	95.00
Custom enclosure	100.00	1	100.00
Total			**1759.00**

Total Estimated Cost: GH¢ 1759.00

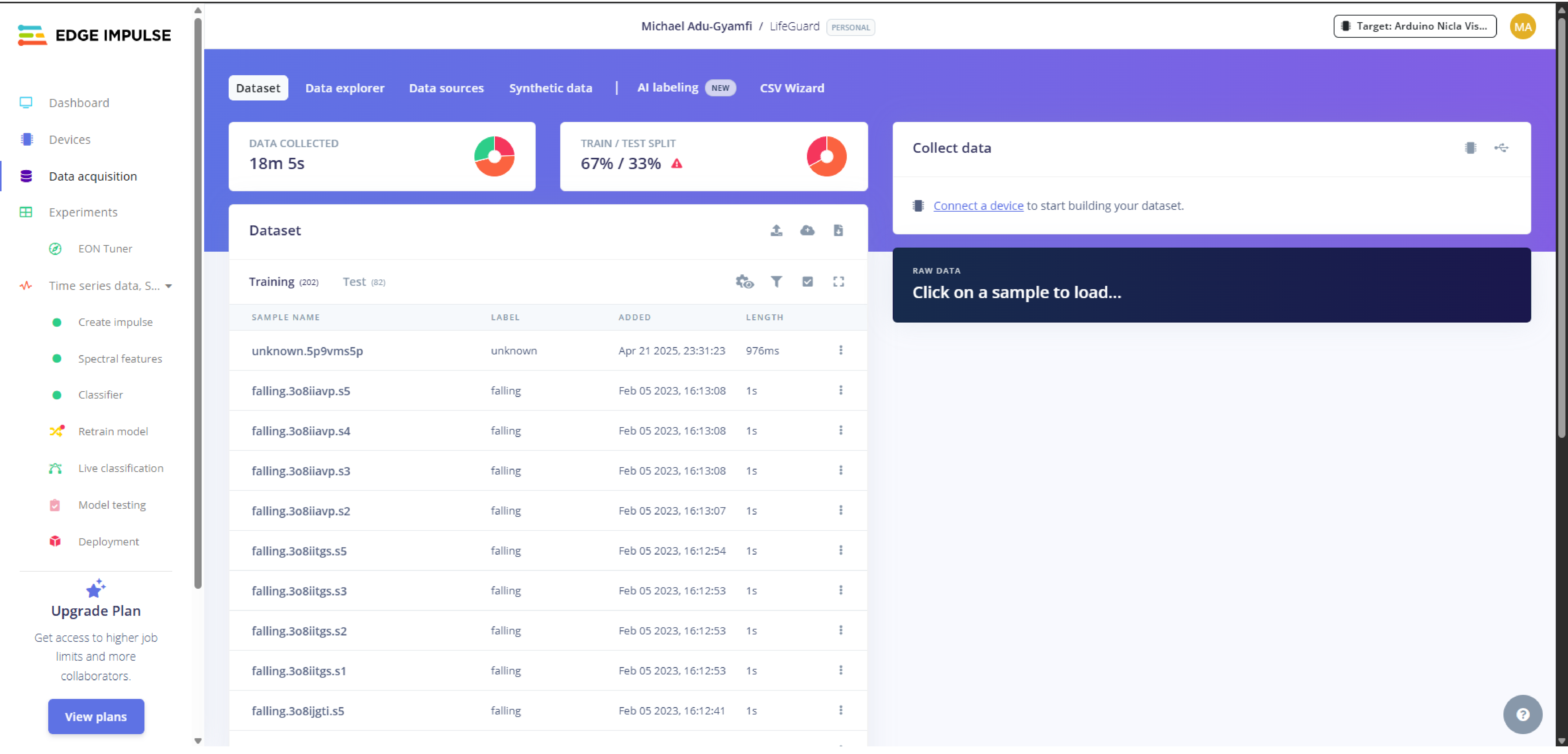
Device Dashboard: Live Sensor Data test & Controls



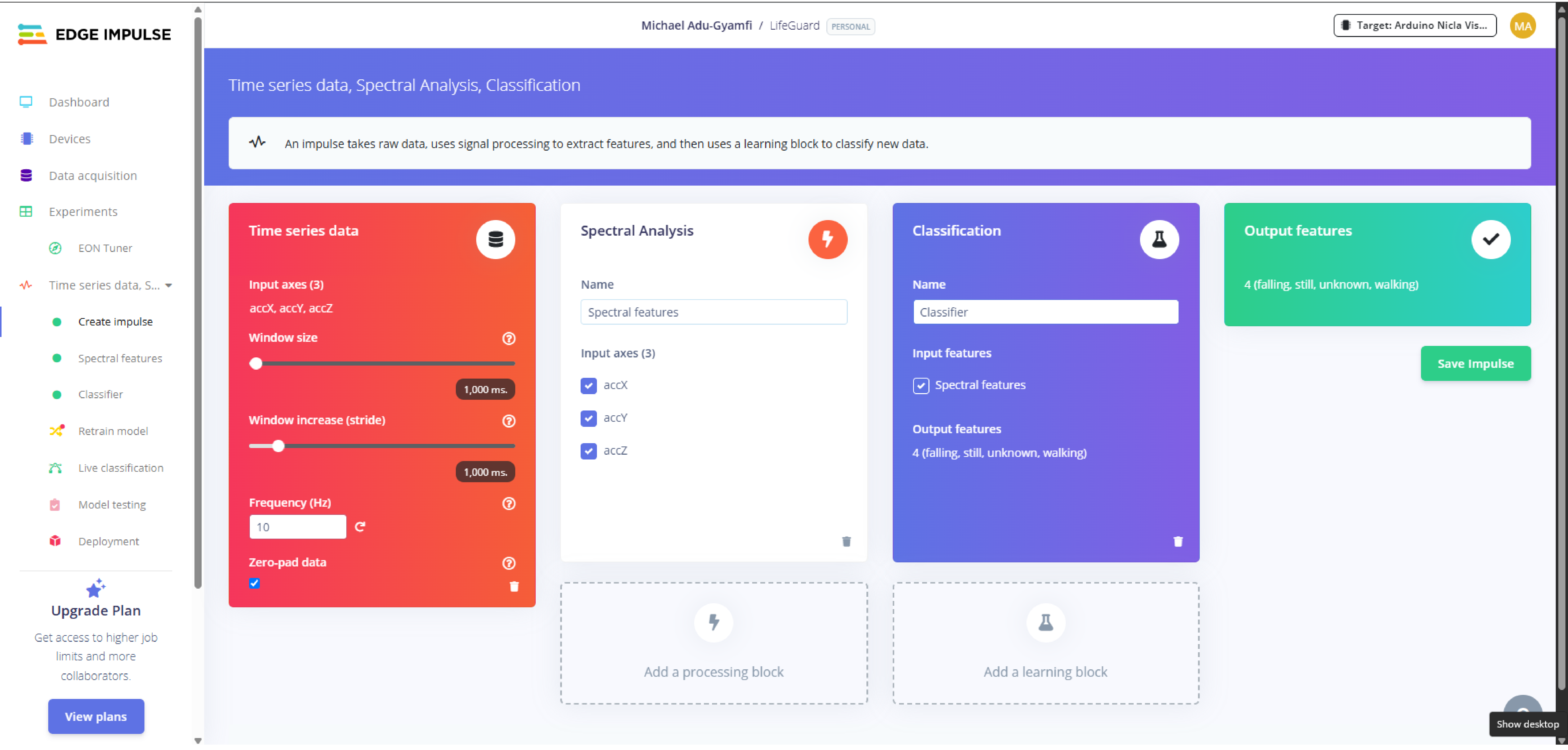
IMPULSE DESIGN: SENSOR DATA & FEATURE EXTRACTION



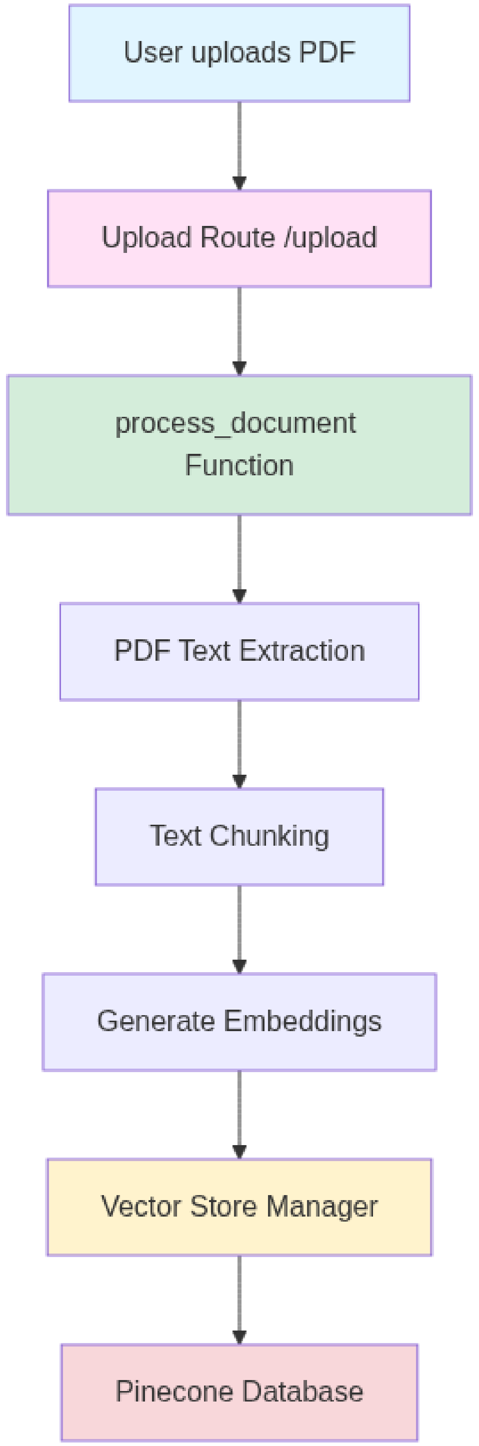
DATASET OVERVIEW & ACQUISITION



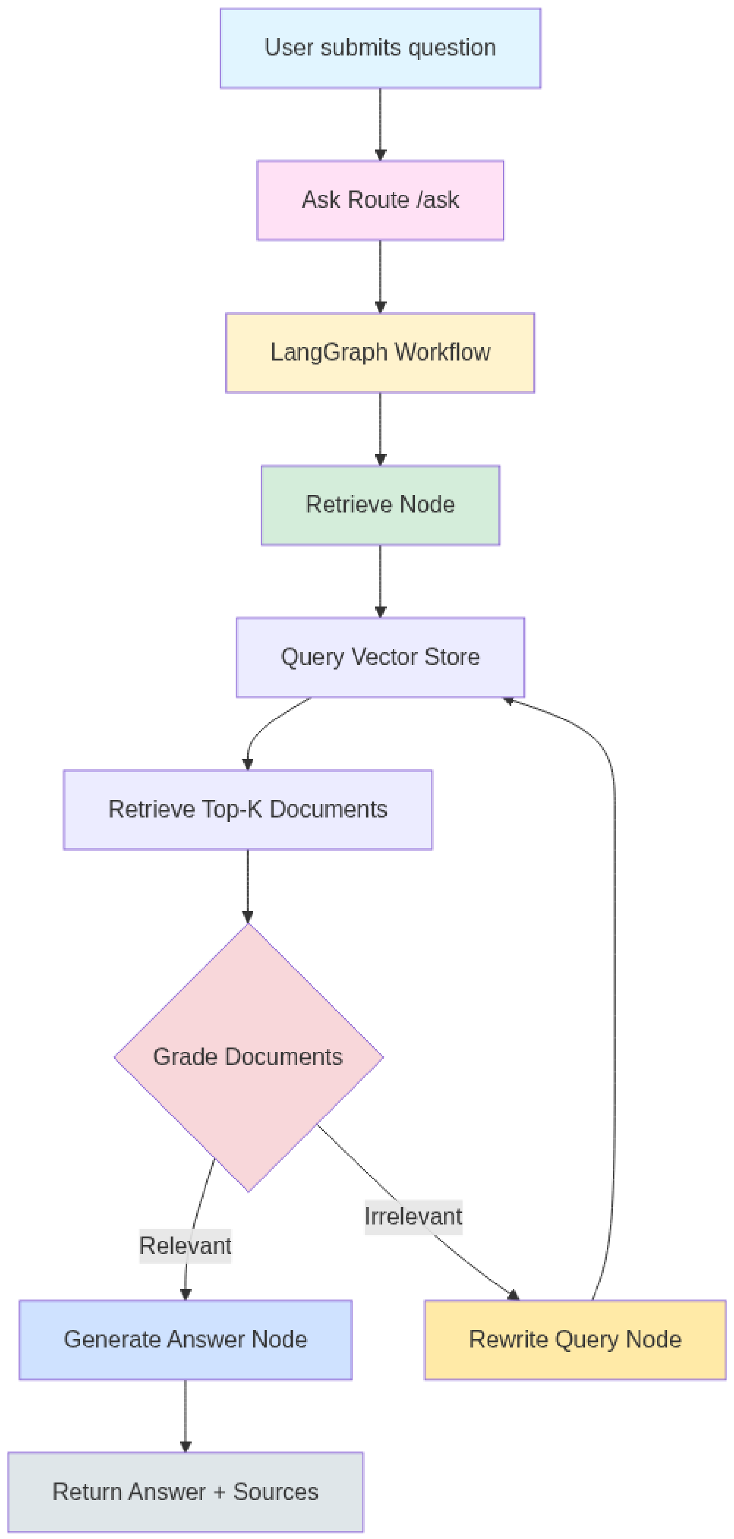
DATASET MANAGEMENT & EXTRACTION



RAG PIPELINE – DOCUMENT UPLOAD FLOW



RAG PIPELINE – QUESTION ANSWERING FLOW



PROJECT DEMO

Personal | LifeGuard

https://lifeguard...

Features | How It Works | Benefits | Download | Log In | Get Started

Your Personal Health Guardian

Monitor your health, track environmental conditions, and stay connected with emergency services - all in one integrated platform.

Get Started →

Star on GitHub

Implementation Timeline

ACTIVITIES	NOV	DEC	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG
Project Planning & Requirement Gathering										
Hardware & Sensor Prototyping – **Arduino Nicla Sense ME**										
Backend & Cloud Development										
Web & Mobile App Development (Flutter)										
Model Training using TinyML on Edge Impulse										
Integration & Testing										
Feedback and Iterations										
Final Deployment and Documentations										

DISCUSSION & CONCLUSION

Discussion

- Proposed system aims to bridge critical gap between health and environmental monitoring
- Integration of Nicla Sense ME board offers comprehensive sensor capabilities at reasonable cost
- Client-Server architecture enables scalable solution for various user groups

Conclusion

- LifeGuard represents innovative approach to integrated safety monitoring
- Potential impact spans healthcare, industrial safety, and personal wellness
- Project lays foundation for future development in wearable safety technology

FUTURE SCOPE

- **Enhanced Sensor Integration**
  - Investigate additional environmental and health-related sensors to expand data capture capabilities
  - Charging system for 3.7v LiPo battery
  - LCD Screen to display critical data

- **Autonomous Emergency Response & Activity Tracking**
  - Implement a fully automated emergency alert system, triggered by on-device machine learning inference (e.g., fall detection or detection of abnormal vital signs)
  - Optimize the firmware to support continuous, low-latency recognition of user activity and real-time data reporting

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THANK YOU!